Customer Churn Analysis

Project Summary | Excel • SQL Server • Python • Power BI

1. Project Overview

This project analyses customer churn for a telecommunications business using the IBM Telco Customer Churn dataset. The goal is to understand the scale of churn, identify customer groups with higher churn rates, and communicate findings through SQL analysis, Python exploratory data analysis (EDA), and an interactive Power BI dashboard.

2. Business Problem

Customer churn reduces recurring revenue and makes customer retention difficult. The business needs a clear view of how churn varies across customer contracts, payment methods, internet services, tenure, and customer characteristics. This analysis describes patterns and associations in the available data; it does not establish that any one factor causes churn.

3. Dataset and Preparation

Dataset: IBM Telco Customer Churn. The dataset contains 7,043 customer records and 21 columns. The data was checked for duplicate records and customer IDs. Eleven whitespace-only TotalCharges entries, associated with zero-tenure customers, were converted to numeric zero so the field could be analysed consistently. The cleaned file is named Telco_Customer_Churn_Cleaned.csv.

4. Technology Stack and Workflow

Excel: initial inspection, data-quality checks, and cleaning of TotalCharges.
SQL Server: data storage, validation, aggregations, and churn analysis by customer attributes.
Python: Pandas and NumPy for analysis; Matplotlib and Seaborn for visualisations in a Jupyter Notebook.
Power BI: interactive dashboard with KPI cards, seven churn-analysis charts, and dropdown slicers.

5. Key Project Metrics

Metric	Result
Total customers	7,043
Churned customers	1,869
Retained customers	5,174
Overall churn rate	26.54%

6. Key Findings

Contract: Month-to-month customers have the highest churn rate (42.71%), compared with one-year (11.27%) and two-year contracts (2.83%).

Payment method: Electronic-check customers have the highest churn rate (45.29%). Mailed check is 19.11%, bank transfer automatic is 16.71%, and credit card automatic is 15.24%.

Internet service: Fiber-optic customers show 41.89% churn, compared with DSL at 18.96% and customers with no internet service at 7.40%.

Tenure: Churn is highest among customers with 0–12 months of tenure (47.44%) and decreases across longer-tenure groups: 13–24 months (28.71%), 25–48 months (20.39%), and 49+ months (9.51%).

Senior citizen: The churn rate is 41.68% for senior citizens and 23.61% for non-senior citizens.

Monthly charges: Average monthly charges are $74.44 for churned customers and $61.27 for retained customers.

Customer segment: The segment churn rates are New Customers (0–12 months): 47.44%; Growing Customers (13–48 months): 23.64%; Loyal Customers (49+ months): 9.51%.

7. Power BI Dashboard

The dashboard contains four KPI cards: Total Customers, Churned Customers, Retained Customers, and Overall Churn Rate. Seven stacked column charts analyse churn by contract, payment method, internet service, gender, tenure group, senior-citizen category, and customer segment. Dropdown slicers for Contract, Internet Service, and Gender allow users to explore filtered results.

8. Conclusion

The analysis identifies month-to-month contracts, shorter tenure, electronic-check payments, and fiber-optic internet service as customer groups with comparatively higher observed churn rates. These findings can help the business prioritise further investigation and retention experiments. Any proposed intervention should be tested and evaluated using subsequent customer data.

Note: Results are descriptive and depend on the supplied dataset and the cleaning decisions documented for this project.